Departement Elektriese, Elektroniese en Rekenaar-Ingenieurswese	Department of Electrical, Electronic and Computer Engineering
Eksamen	**Examination**
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Module EBN122	**Module EBN122**
Elektrisiteit en Elektronika	**Electricity and Electronics**
November 2016	November 2016

Eksamensinligting: ***Exam information:***			
Maksimum punte: *Maximum marks:*	90	Volpunte: *Full marks:*	94
Duur van vraestel: *Duration of paper:*	180 minute *180 minutes*	Oopboek / toeboek: *Open / closed book:*	Toe *Closed*
Totale aantal bladsye (hierdie blad ingesluit): *Total number of pages (including this page):*		27 (Antwoordblad uigesluit / *Answer sheet excluded)*	

BELANGRIK- ***IMPORTANT***

1. *The examination regulations of the University of Pretoria apply.*
   Die eksamenregulasies van die Universiteit van Pretoria geld.
2. *All questions must be answered on the ANSWER SHEET.*
   Alle vrae moet op die ANTWOORDBLAD beantwoord word.
3. *The English text of this question paper will be given precedence in the interpretation of the content.*
   Die Engelse teks van hierdie vraestel sal voorkeur geniet met betrekking tot die interpretasie van die inhoud.
4. *Students may do their own rough calculations on the question paper – the question paper will not be taken in.*
   Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
5. *Answers must include units!*
   Antwoorde moet eenhede insluit!
6. *Only non-programmable calculators may be used.*
   Slegs nie-programmeerbare sakrekenaars mag gebruik word.
7. ***Polar form:*** $A\angle\theta$ ***; Rectangular form:*** $a + jb$
   Polêre vorm: $A\angle\theta$ **; Reghoekige vorm:** $a + jb$
8. ***Final answers must be written as real numbers, and not as fractions.***
   Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.

Interne Eksaminator(e): *Internal examiner(s):*	D. le Roux, J. Joubert, R. Dos Santos, D. Ramotsoela
Eksterne eksaminator: External examinor:	W. Odendaal

QUESTION/VRAAG 1 [18]

Question 1.1 [6]

The voltage and current at the terminals of the circuit below are zero for $t < 0$ and $t > 40$s. In the interval between 0 and 40s the expressions are

$$v(t) = t(1 - 0.025t) \text{ V}$$

$$i(t) = 4 - 0.2t \text{ A}$$

(a) Calculate the power associated with the element at $t = 35$s.
(b) Calculate the amount of charge entering the element during the interval between 0 and 20s.
(c) Calculate the total energy delivered to the circuit during the interval between 0 and 40s.

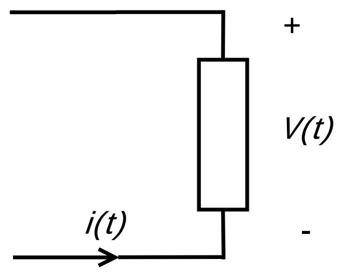
Vraag 1.1 [6]

Die spanning en stroom by die terminale van die stroombaan hieronder is nul vir $t < 0$ en $t > 40$s. In die interval tussen 0 en 40s is die uitdrukkings

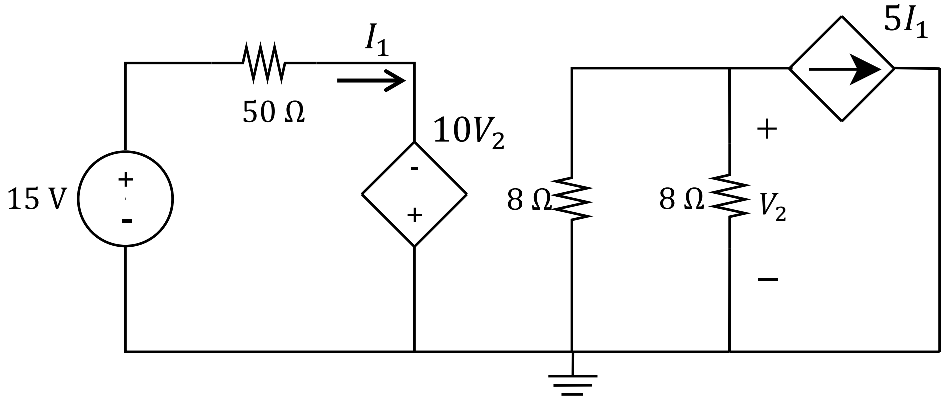
$$v(t) = t(1 - 0.025t) \text{ V}$$

$$i(t) = 4 - 0.2t \text{ A}$$

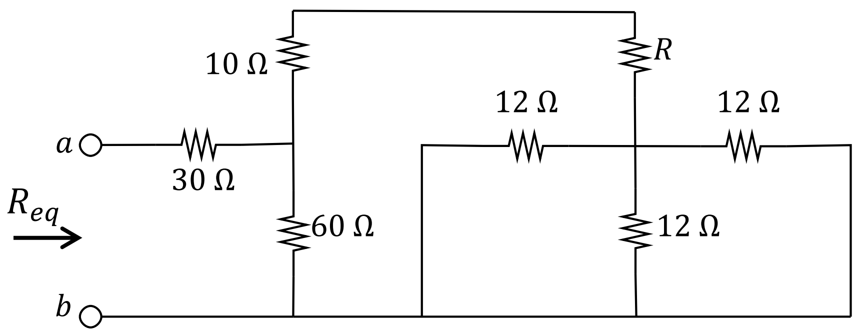
(a) Bereken die drywing ge-assosieer met die element by $t = 35$s.
(b) Bereken die aantal lading wat die element binnevloei in die tydperk tussen 0 en 20s.
(c) Bereken die totale energie gelewer aan die stroombaan in die tydperk tussen 0 en 40s.



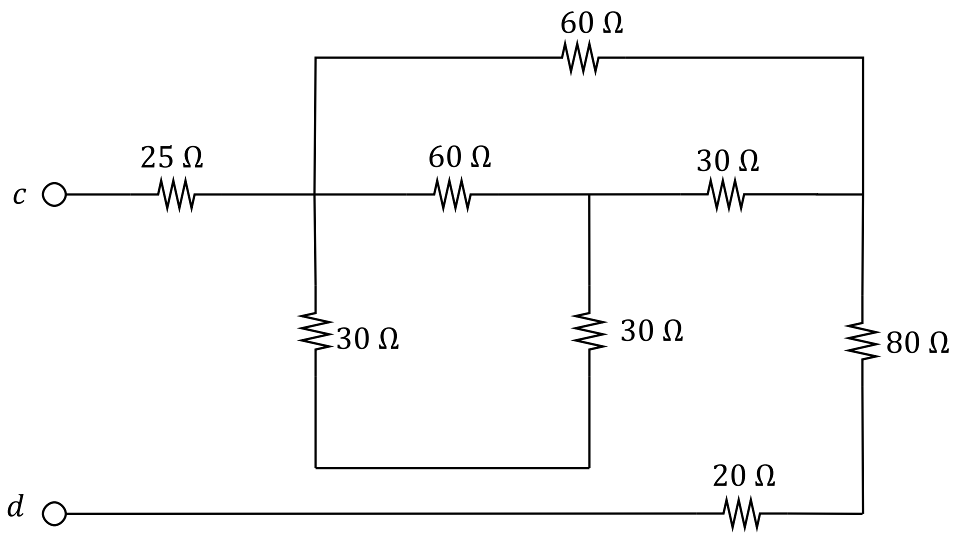
Question 1.2 [4] Determine I_1 and V_2.	**Vraag 1.2 [4]** Bepaal I_1 en V_2.



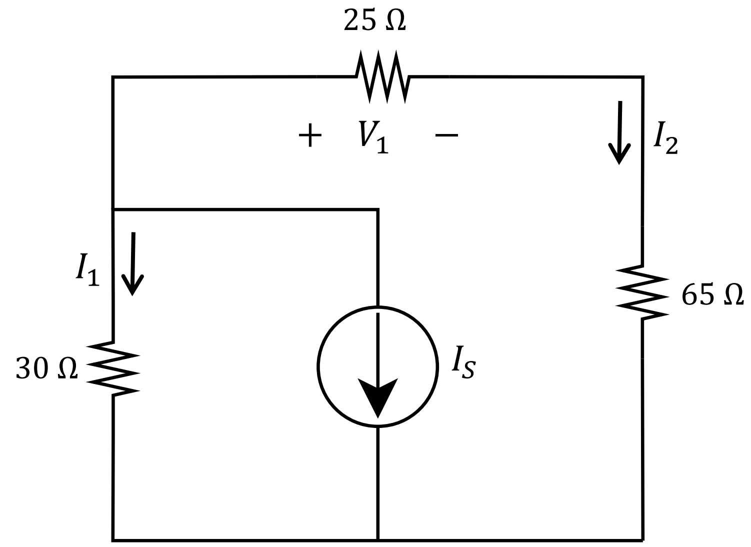
Question 1.3 [**2**]	**Vraag 1.3** [**2**]
Determine R if $R_{eq} = 50\ \Omega$.	Bereken R indien $R_{eq} = 50\ \Omega$.



Question 1.4 [2]	Vraag 1.4 [2]
Calculate the equivalent resistance w.r.t terminals *c*-*d*.	Bereken die ekwivalente weerstand m.b.t. terminale *c*-*d*.

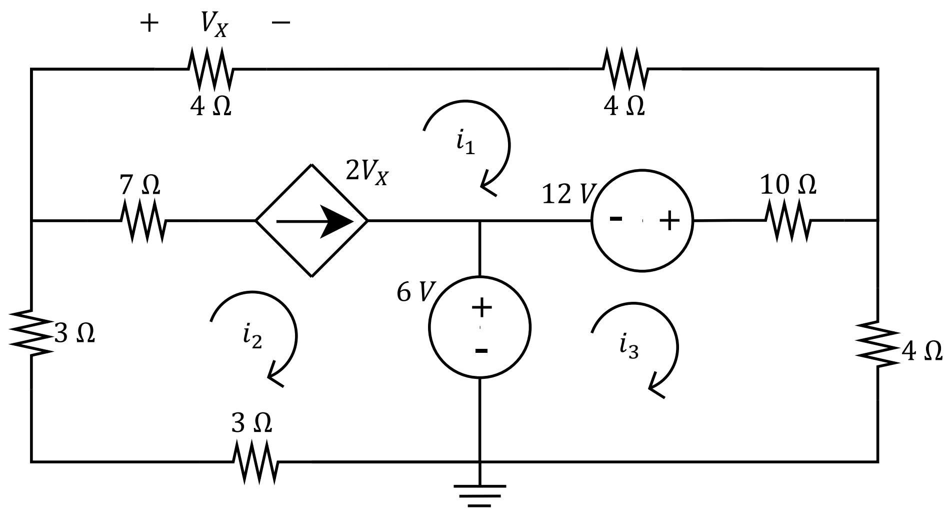


Question 1.5 [4]	**Vraag 1.5 [4]**
For the circuit below, calculate a) I_S if I_2 = 137.5 mA b) V_1 if I_1 = 225 mA	Vir onderstaande kringbaan, bereken a) I_S indien I_2 = 137.5 mA b) V_1 indien I_1 = 225 mA

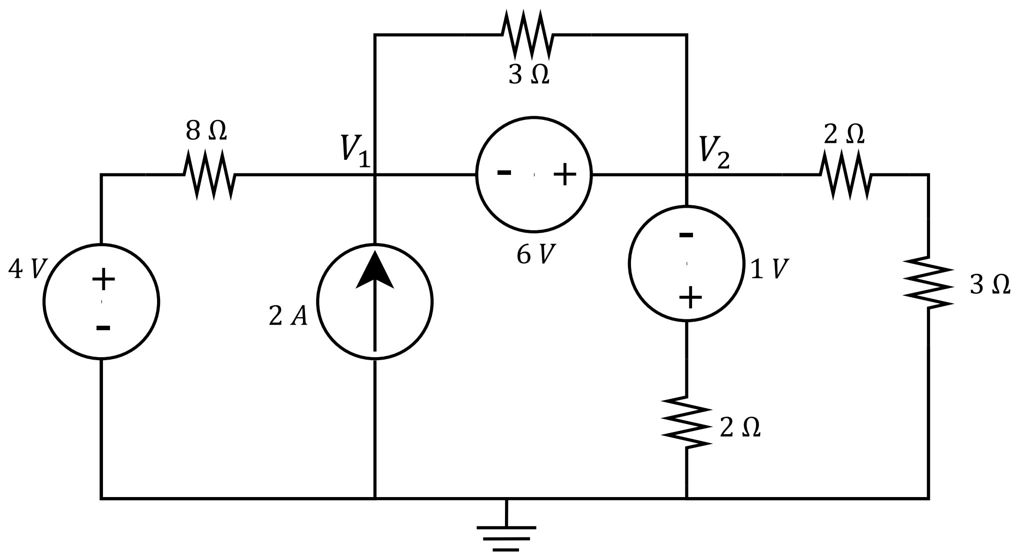


QUESTION/VRAAG 2 [32]

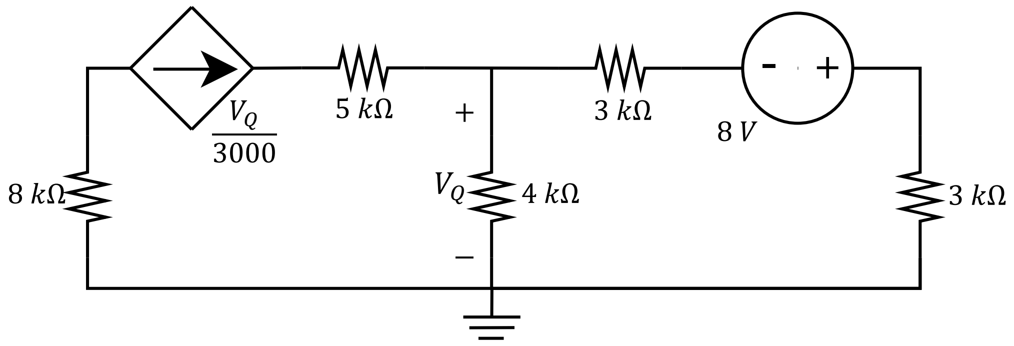
Question 2.1 [4] Use mesh analysis to find the missing coefficients for the following equations: $$ai_1 + bi_2 + ci_3 = -18$$ $$di_1 - i_2 = 0$$ $$-10i_1 + 14i_3 = 18$$	**Vraag 2.1 [4]** Gebruik lus-analise om die onbekende koeffisiente te bepaal in die volgende vergelykings: $$ai_1 + bi_2 + ci_3 = -18$$ $$di_1 - i_2 = 0$$ $$-10i_1 + 14i_3 = 18$$



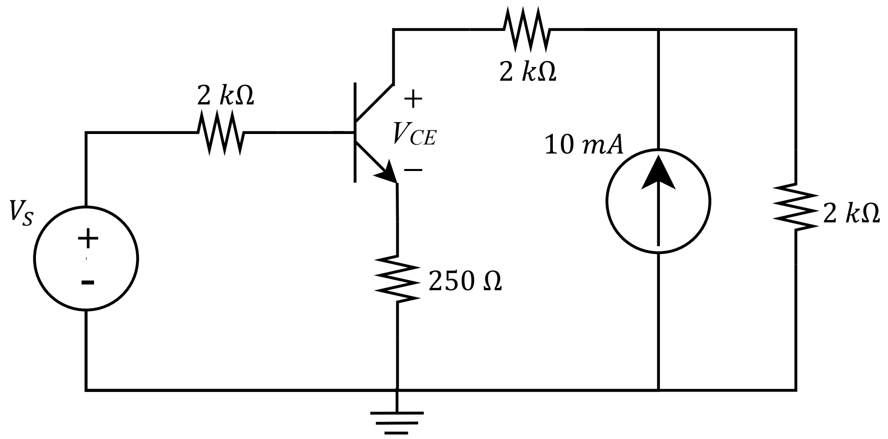
Question 2.2 [4] Use nodal analysis to set up two equations of the following form: $eV_1 + fV_2 = 6$ $gV_1 + hV_2 = 80$	**Vraag 2.2 [4]** Gebruik node-analise om twee vergelykings van die volgende vorm op te stel: $eV_1 + fV_2 = 6$ $gV_1 + hV_2 = 80$



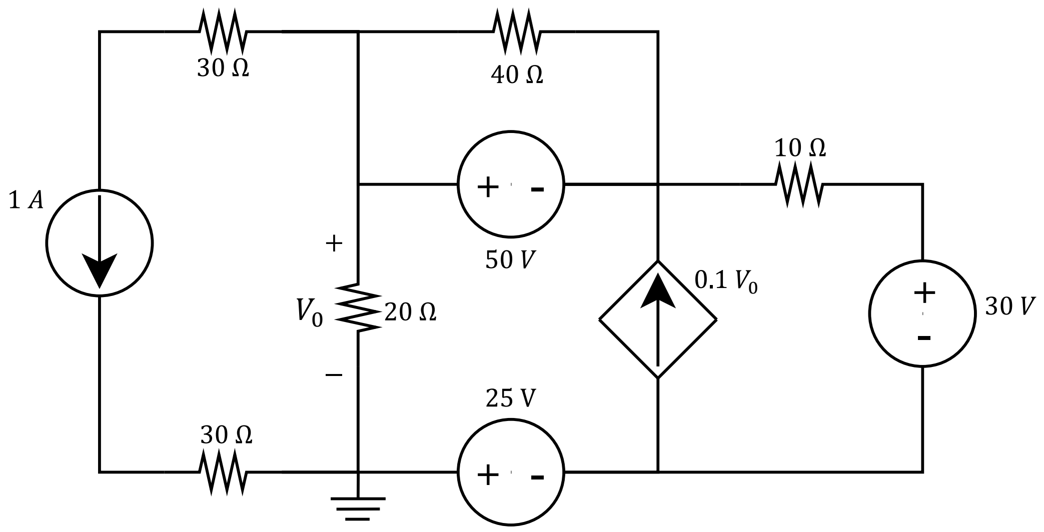
Question 2.3 [2]	Vraag 2.3 [2]
Determine V_Q. Use nodal or mesh analysis.	Bepaal V_Q. Gebruik node- of lusanalise.



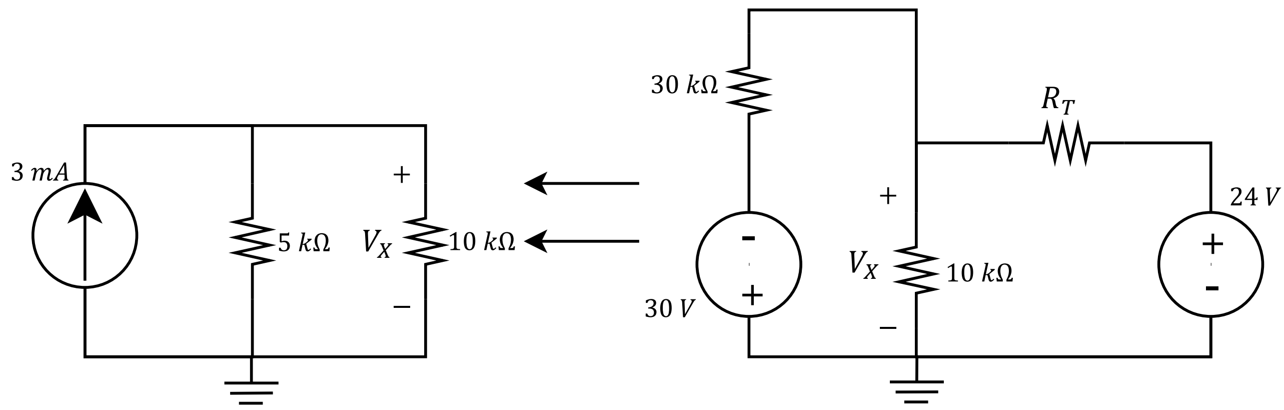
Question 2.4 [4] Given: $\beta = 99$ and $V_{BE} = 0.7$ V. a) If $V_S = 3.4$ V, determine I_B. b) If $I_B = 20$ μA, determine V_{CE}.	**Vraag 2.4 [4]** Gegee: $\beta = 99$ en $V_{BE} = 0.7$ V. a) Indien $V_S = 3.4$ V, bepaal I_B. b) Indien $I_B = 20$ μA, bepaal V_{CE}.



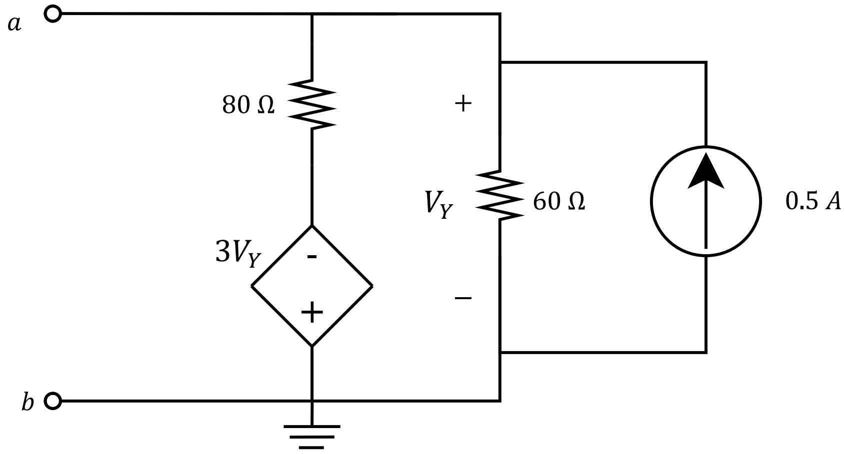
Question 2.5 [4] Use superposition to determine the unique contribution of a) the 1 A current source to V_0, and b) the 30 V voltage source to V_0.	**Vraag 2.5 [4]** Gebruik superposisie om die unieke bydra van a) die 1 A stroombron tot V_0, b) die 30 V spanningsbrong tot V_0, te bereken.



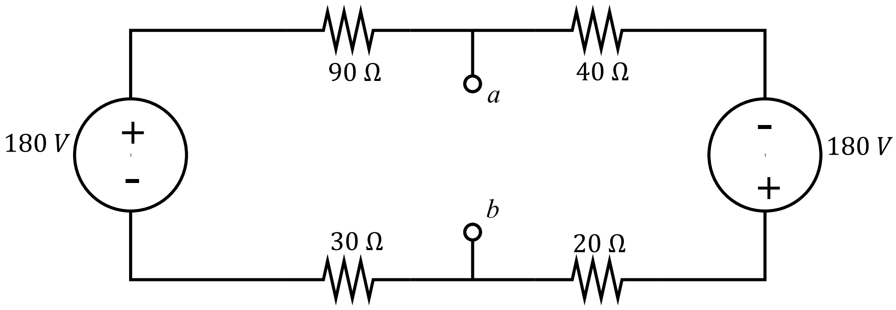
Question 2.6 [4] The circuit on the left was obtained through source transformation of the circuit on the right. a) Calculate the value of R_T. b) Calculate the value of V_X.	**Vraag 2.6 [4]** Die linkerkantse kringbaan is verkry deur brontransformasie van die regterkantse kringbaan. a) Bereken die waarde van R_T. b) Bereken die waarde van V_X.



Question 2.7 [4] Calculate the Thevenin equivalent with respect to terminals *a*-*b*.	**Vraag 2.7 [4]** Bereken die Thevenin ekwivalent met verwysing tot terminale *a* en *b*.



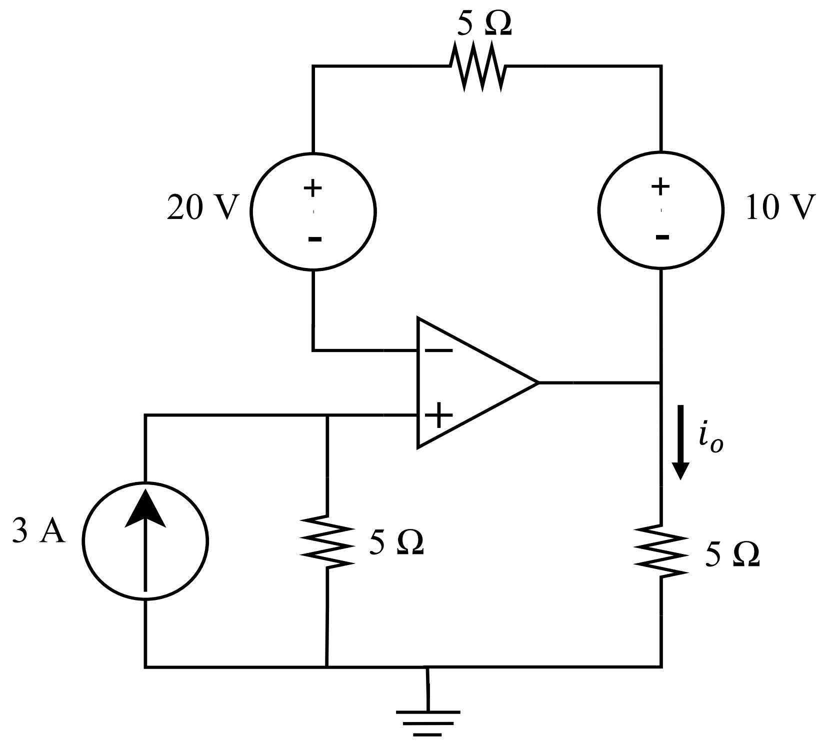
Question 2.8 [4] Determine the Norton equivalent with respect to terminals *a* and *b*.	**Vraag 2.8 [4]** Bepaal die Norton ekwivalent met betrekking tot terminale *a* en *b*.



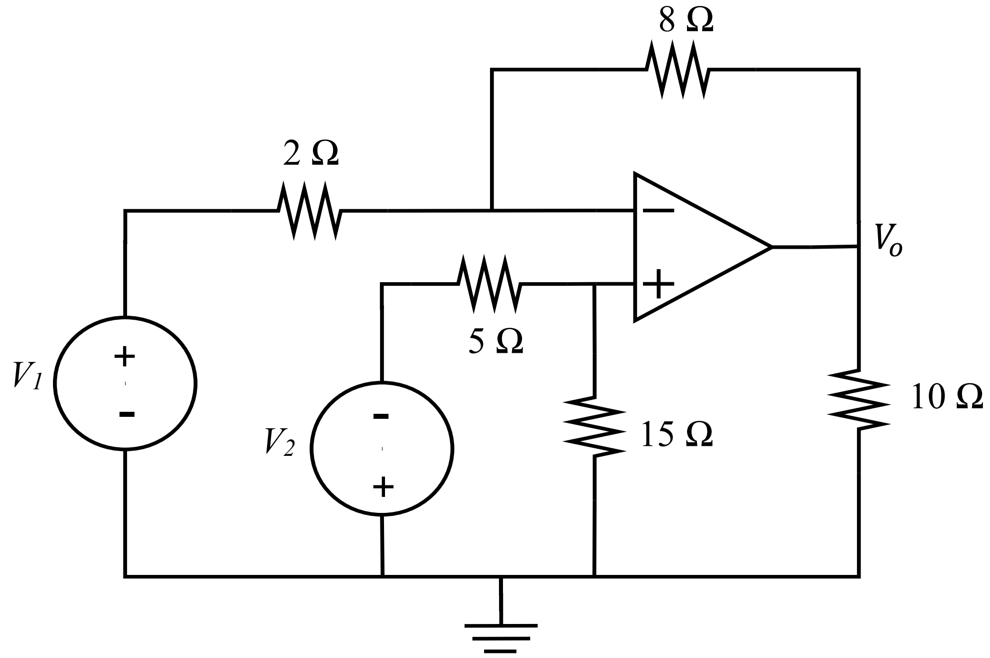
Question 2.9 [2]	**Vraag 2.9 [2]**
A circuit delivers maximum power if the load connected to the circuit is 4 Ω. If the load is changed to 2 Ω, the power delivered to the load is 8 W. Calculate the Thevenin equivalent voltage of the circuit.	‘n Kringbaan lewer maksimum drywing indien daar ‘n las van 4 Ω aan die kringbaan gekoppel is. Indien die las na 2 Ω verander word, verbuik die las 8 W. Bereken die Thevenin ekwivalente spanning van die kringbaan.

VRAAG/QUESTION 3 [20]

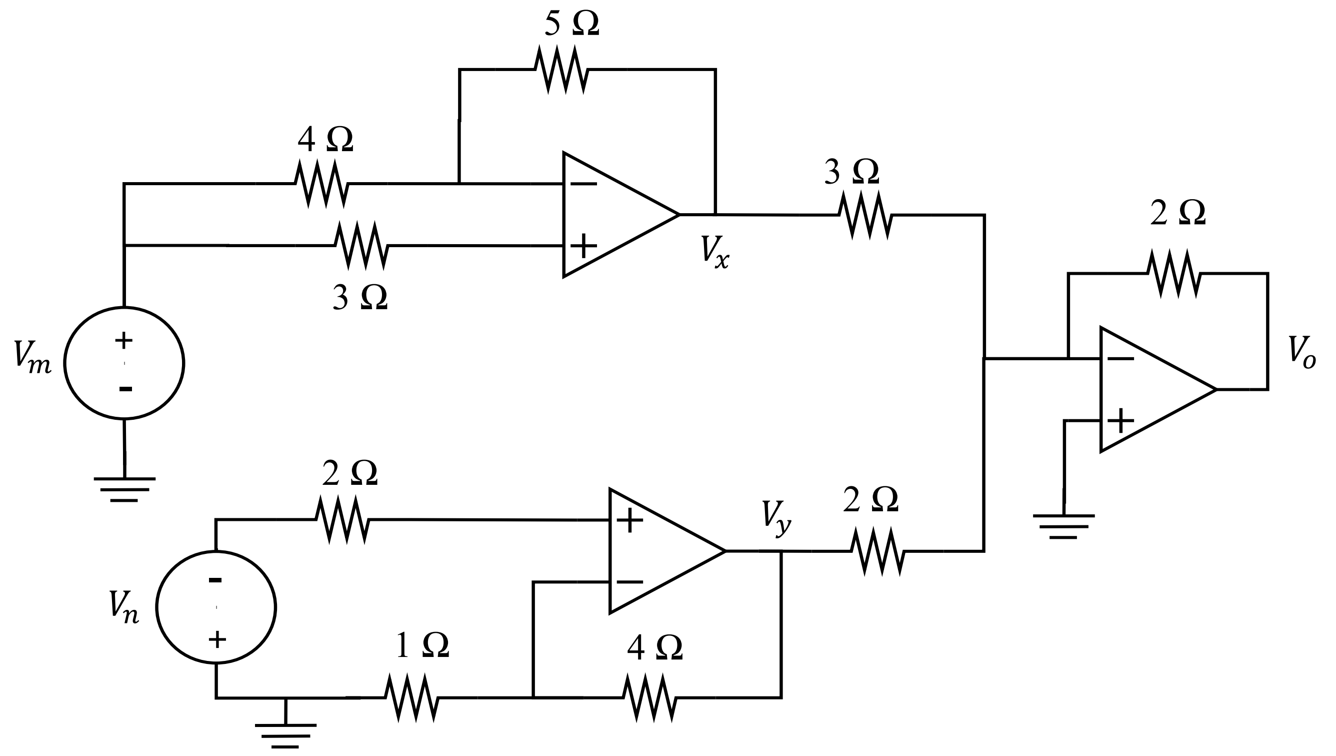
Question 3.1 [2]	Vraag 3.1 [2]
Calculate i_o.	Bereken i_o.



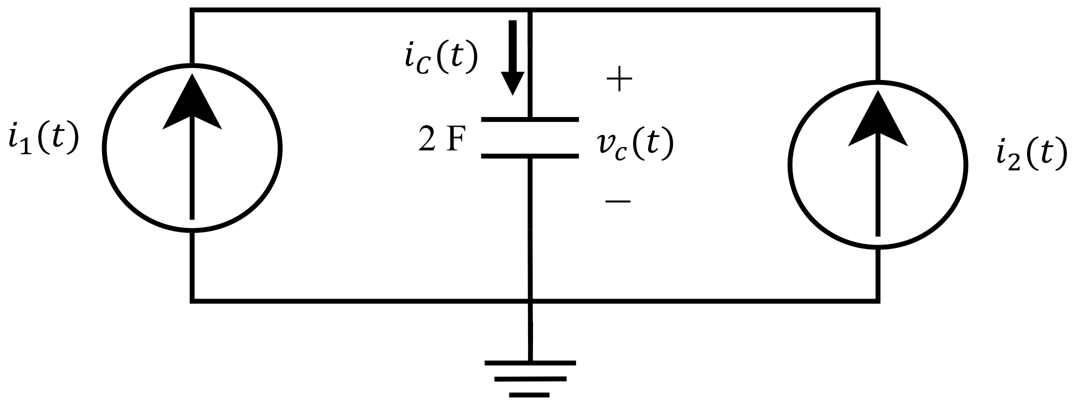
Question 3.2 [4] Calculate V_o, a) if $V_1 = 4\text{V}$ and $V_2 = 0\text{V}$ b) if $V_1 = 6\text{V}$ and $V_2 = 4\text{V}$	**Vraag 3.2 [4]** Bereken V_o, a) as $V_1 = 4\text{V}$ en $V_2 = 0\text{V}$ b) as $V_1 = 6\text{V}$ en $V_2 = 4\text{V}$



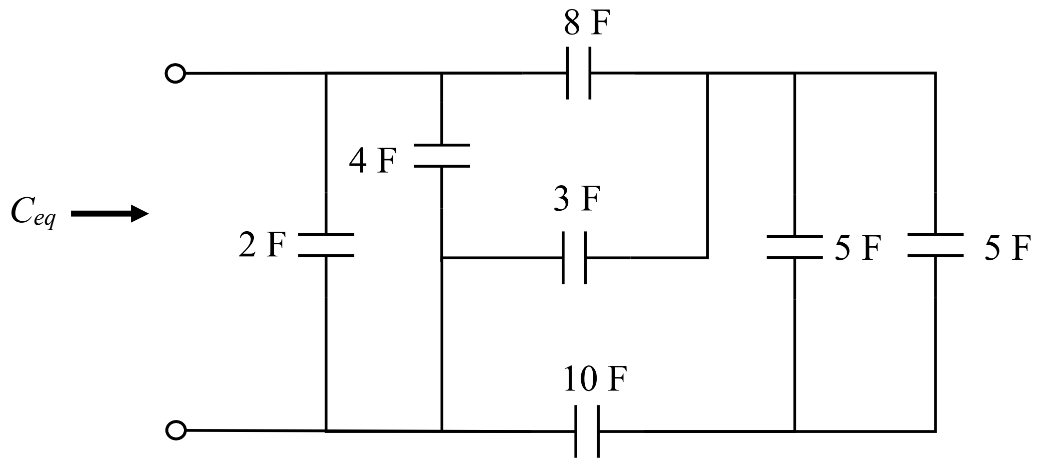
Question 3.3 [6]	**Vraag 3.3 [6]**
If $V_x = 6$ V and $V_y = -10$ V, then determine: a) V_m b) V_n c) V_o	As $V_x = 6$ V en $V_y = -10$ V, dan bepaal: a) V_m b) V_n c) V_o



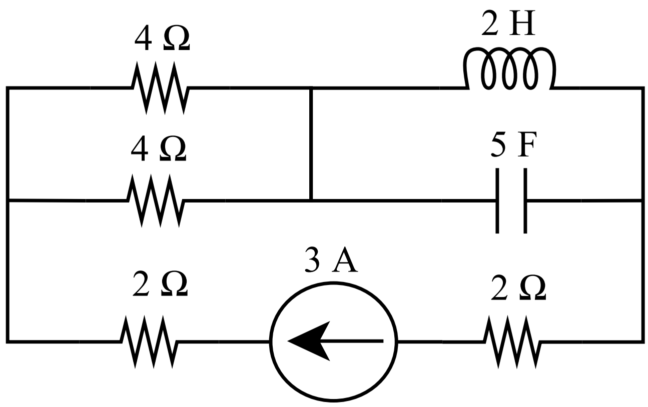
Question 3.4 [2]	Vraag 3.4 [2]
Calculate $v_C(t)$ at $t = 2$, if $v_C(0) = 0\text{V}$ and, $i_1(t) = 2e^{-4t}$ A $i_2(t) = 8t$ A	Bepaal $v_C(t)$ vir $t = 2$, as $v_C(0) = 0\text{V}$ en, $i_1(t) = 2e^{-4t}$ A $i_2(t) = 8t$ A



Question 3.5 [2]	Vraag 3.5 [2]
Calculate C_{eq}.	Bepaal C_{eq}.



Question 3.6 [4] Calculate the energy stored by the capacitor (w_C) and inductor (w_L). Assume direct current conditions.	**Vraag 3.6 [4]** Bepaal die energie gestor in die kapasitor (w_C) en die inductor (w_L). Aanvaar gelykstroom toestande.

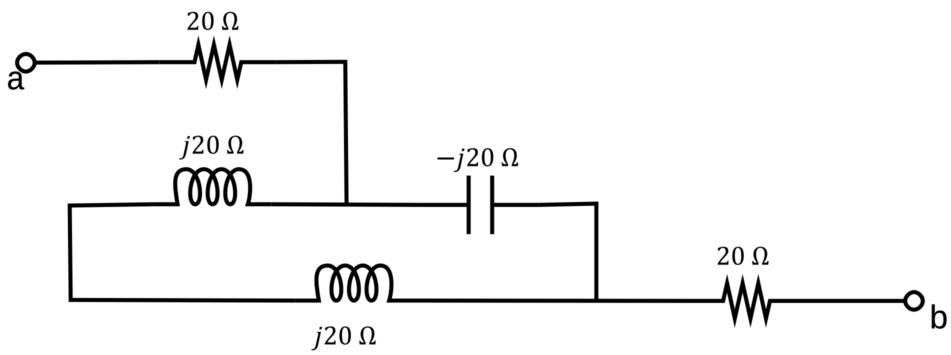


VRAAG/QUESTION 4 [24]

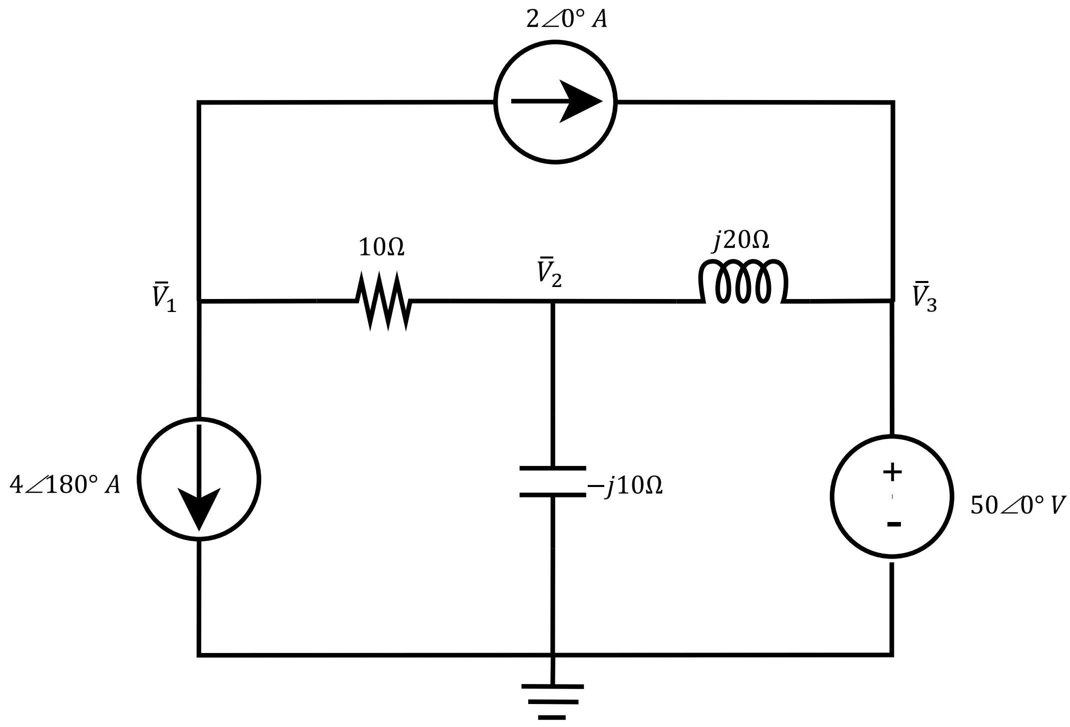
Question 4.1 [6]	**Vraag 4.1 [6]**
a) Write $v(t)$ as a single phasor in **rectangular form**: $v(t) = 15cos(\omega t + 45°) - 10sin(\omega t + 45°)$ V	a) Skryf $v(t)$ as 'n enkele fasor in **reghoekige vorm**: $v(t) = 15cos(\omega t + 45°) - 10sin(\omega t + 45°)$ V
b) What is the voltage across a 2 μF capacitor when the current through it is $i_C(t) = 4sin(10^6 t + 25°)$ A. Assume the current is flowing into the + of the reference voltage defined accross the capacitor. Assume $v_C(0) = 0$ V.	b) Bepaal die spanning oor 'n 2 μF kapasitor as die stroom deur die kapasitor $i_C(t) = 4sin(10^6 t + 25°)$ A is. Aanvaar dat die stroom invloei by die + van die verwysings-spanning wat gedefineer is oor die kapasitor. Aanvaar $v_C(0) = 0$ V.
c) A current source of $i(t) = 10sin(377t + 30°)$ A is applied to a single component. The resulting voltage across the element is $v(t) = -65cos(377t + 120°)$ V. Determine the value of the component (you have to decide if it is a resistor, inductor or capacitor).	c) 'n Stroombron $i(t) = 10sin(377t + 30°)$ A word aangewend oor 'n enkele komponent. Die resultante spanning oor die element is $v(t) = -65cos(377t + 120°)$ V. Bereken die waarde van die komponent (besluit eers of dit 'n weerstand, induktor of kapasitor is).

Question 4.2 [4]	Vraag 4.2 [4]
a) Simplify and write the answer in **polar form**: $\bar{I}_1 = \frac{(3+j4)^* + (8\angle -90°)^*}{2e^{j45°}}$	a) Vereenvoudig en skryf die antwoord in **polêre vorm**: $\bar{I}_1 = \frac{(3+j4)^* + (8\angle -90°)^*}{2e^{j45°}}$
b) Determine $\bar{V}_1$ in **polar form**: $\begin{bmatrix} 2+j4 & -j2 \\ -j2 & 2 \end{bmatrix}\begin{bmatrix} \bar{V}_1 \\ \bar{V}_2 \end{bmatrix} = \begin{bmatrix} 2\angle 180° \\ 2\angle 0° \end{bmatrix}$	b) Bepaal $\bar{V}_1$ in **polêre vorm**: $\begin{bmatrix} 2+j4 & -j2 \\ -j2 & 2 \end{bmatrix}\begin{bmatrix} \bar{V}_1 \\ \bar{V}_2 \end{bmatrix} = \begin{bmatrix} 2\angle 180° \\ 2\angle 0° \end{bmatrix}$

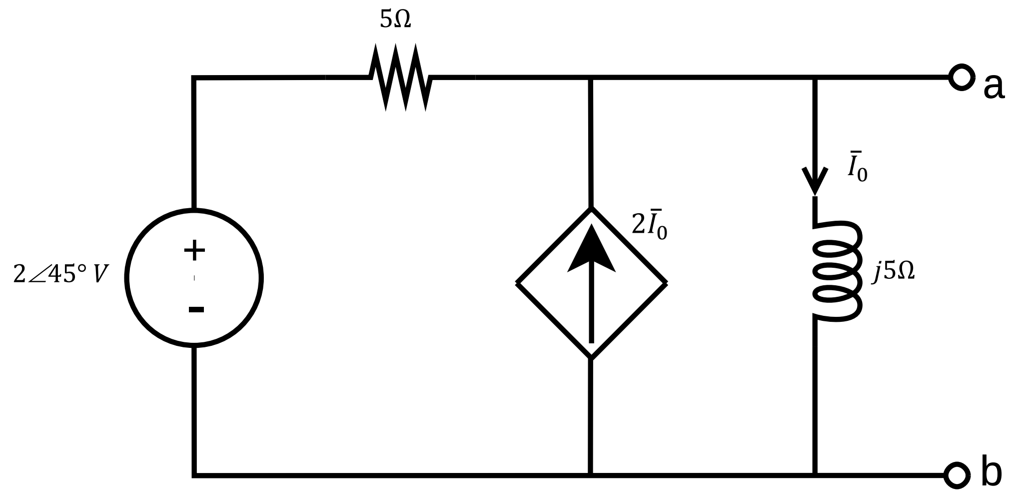
Question 4.3 [2]	Vraag 4.3 [2]
Determine the total impedance $\bar{Z}_{ab}$ between terminals a and b. Write the answer in **rectangular form**.	Bepaal die totale impedansie $\bar{Z}_{ab}$ tussen terminale a en b. Skryf die antwoord in **reghoekige vorm**.



Question 4.4 [4]	**Vraag 4.4 [4]**
Use nodal analysis to set up two equations of the following form: $$(a_1 + jb_1)\bar{V}_1 + (a_2 + jb_2)\bar{V}_2 = 20\angle 0°$$ $$(a_3 + jb_3)\bar{V}_1 + (a_4 + jb_4)\bar{V}_2 = 50\angle -90°$$	Gebruik node-analise om twee vergelykings van die volgende vorm op te stel: $$(a_1 + jb_1)\bar{V}_1 + (a_2 + jb_2)\bar{V}_2 = 20\angle 0°$$ $$(a_3 + jb_3)\bar{V}_1 + (a_4 + jb_4)\bar{V}_2 = 50\angle -90°$$



Question 4.5 [4]	**Vraag 4.5 [4]**
Find the Thevenin equivalent wrt terminals a-b. Write the answer as phasors in **polar form**.	Bepaal die Thevenin-ekwivalent tov terminale a-b. Skryf die antwoorde as fasors in **polêre vorm**.



Question 4.6 [4] Find the Norton equivalent wrt terminals a-b. Write the answers as phasors in **polar form**.	**Vraag 4.6 [4]** Bepaal die Norton-ekwivalent tov terminale a-b. Skryf die antwoorde as fasors in **polêre vorm**.